3D WEBSOFT INC. FOUNDATION

SCHOOL FUTURE TECH PROGRAM

Faculty Name :

Course Duration : 5 Months (150 Days)

Classes : Offline & Online

Daily Teaching : 3 Hours

Total Hours : 300 Hours

LEVEL 3 –Future Engineers(9th–10th Standard)

MODULE 1 – Python Foundations (Classes 1–10)

Class 2 – Introduction to Programming & Python

Complete Theory & Practical Notes

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Learning Objectives:-

At the end of this class, students will be able to:

By the end of this class, students will be able to:

- Explain what programming means.
- Understand how a computer follows instructions.
- Identify programs, programming languages, and Python.
- Understand basic Python syntax.
- Run their first Python program.
- Use print() to display text and numbers.
- Perform simple calculations in Python.
- Create a small "About Me" Python program.

Session Plan

Topic	*Type*
Welcome & Icebreaker	Activity
What is Programming?	Theory
How Computers Understand Instructions	Theory + Activity
Programming Languages & Python	Theory
First Python Program	Practical
Break + Coding Challenge	Activity
`print()` and Python Syntax	Practical
Numbers & Calculations	Practical
Mini Project: About Me	Practical
Quiz + Recap	Assessment

1. WELCOME & ICEBREAKER

Teacher Introduction:-

Welcome, Future Engineers!

Today we are going to teach a computer how to follow our instructions.

Ask students:

- What is a computer?
- What can a computer do?
- Can a computer think by itself?
- How does a computer know what to do?
- Have you ever used an app or computer game?

Then tell them:

Behind every app, website, game, robot and AI system are instructions written by programmers.

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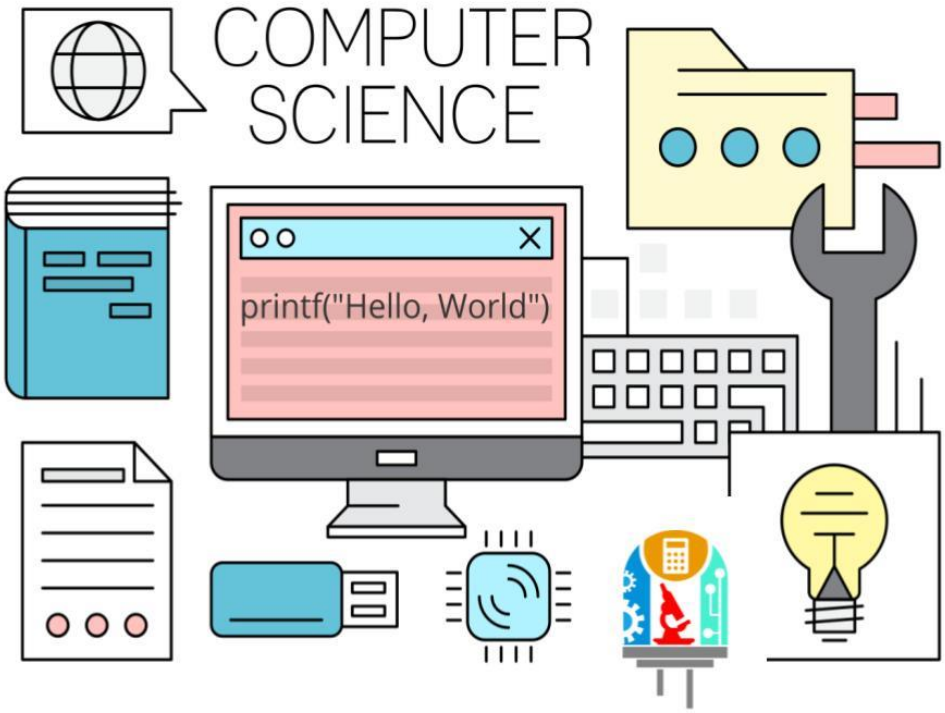

Coding Starts Early

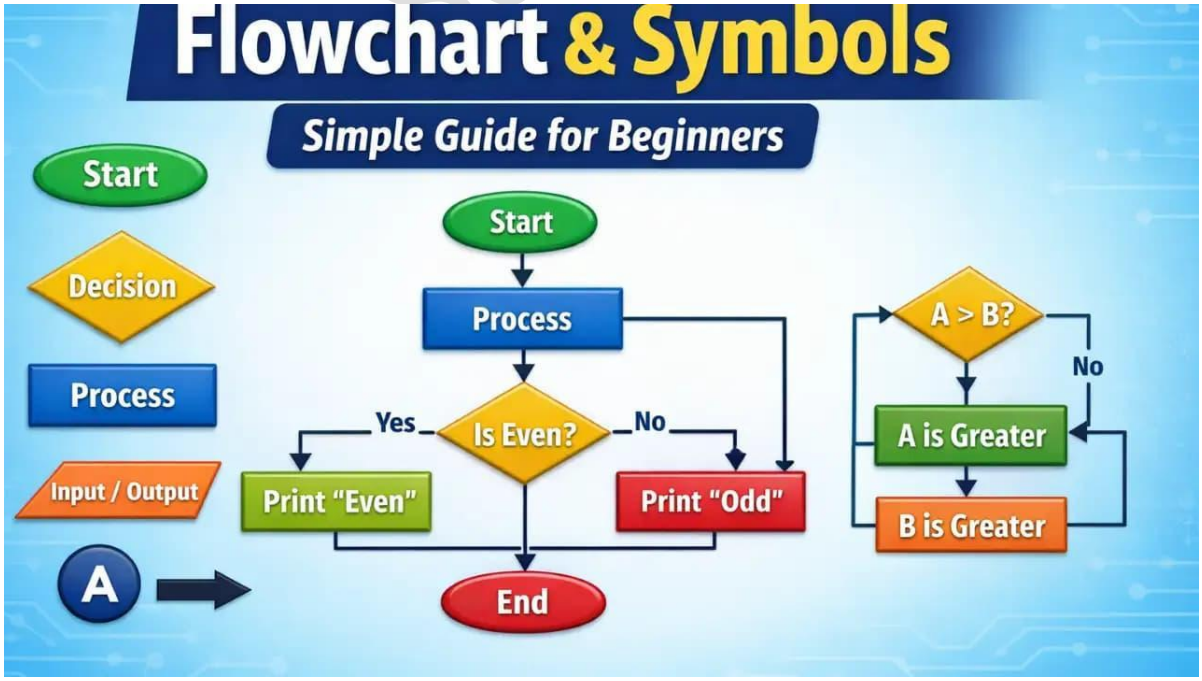


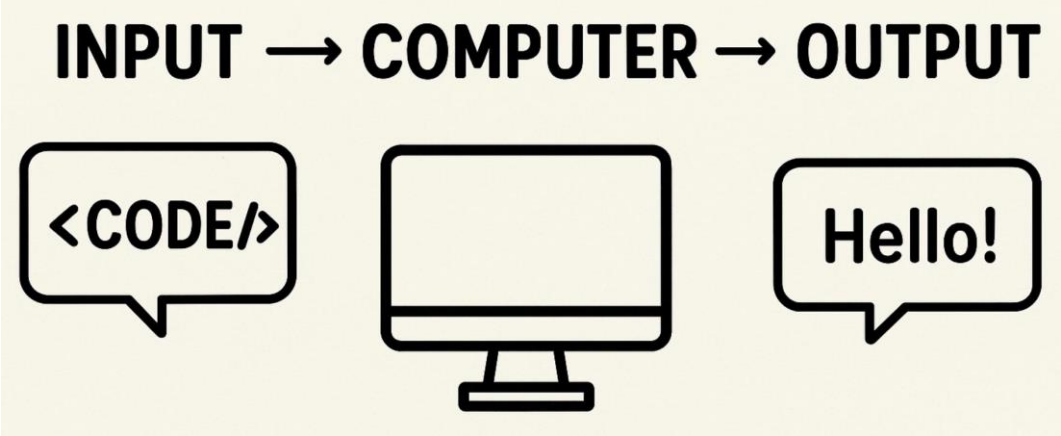


2. WHAT IS PROGRAMMING?

THEORY







Simple Definition

Write on the board:

Programming = Giving instructions to a computer to perform a task.

For example, if we want a computer to calculate:

10 + 20

we give it an instruction.

```
print(10 + 20)
```

The computer produces:

```
30
```

Real-Life Example

How do you make a cup of tea?

Students may say:

1. Take a cup.
2. Add tea.
3. Add hot water.
4. Add milk.
5. Add sugar.
6. Mix.

A computer also needs instructions in the correct order.

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ACTIVITY: HUMAN ROBOT

Choose one student to be the **robot**.

Another student gives instructions.

Example:

```
1. Stand up.
2. Turn right.
3. Walk 3 steps.
4. Pick up the pencil.
5. Come back.
```

If the instruction is unclear, the "robot" should not guess.

Computers are similar.

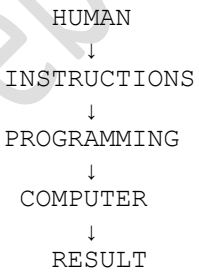
They follow instructions very precisely.

Instruction → Computer → Result

3. HOW COMPUTERS FOLLOW INSTRUCTIONS

THEORY

Draw this on the board:



Example

Human:

"Calculate 5 + 3."

Python:

```
print(5 + 3)
```

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Computer:

```
8
```

Important Concept: Algorithm

Explain:

An algorithm is a step-by-step method for solving a problem.

Example:

Algorithm for brushing teeth

```
1. Take toothbrush
2. Add toothpaste
3. Brush teeth
4. Rinse mouth
5. Clean toothbrush
```

Then explain:

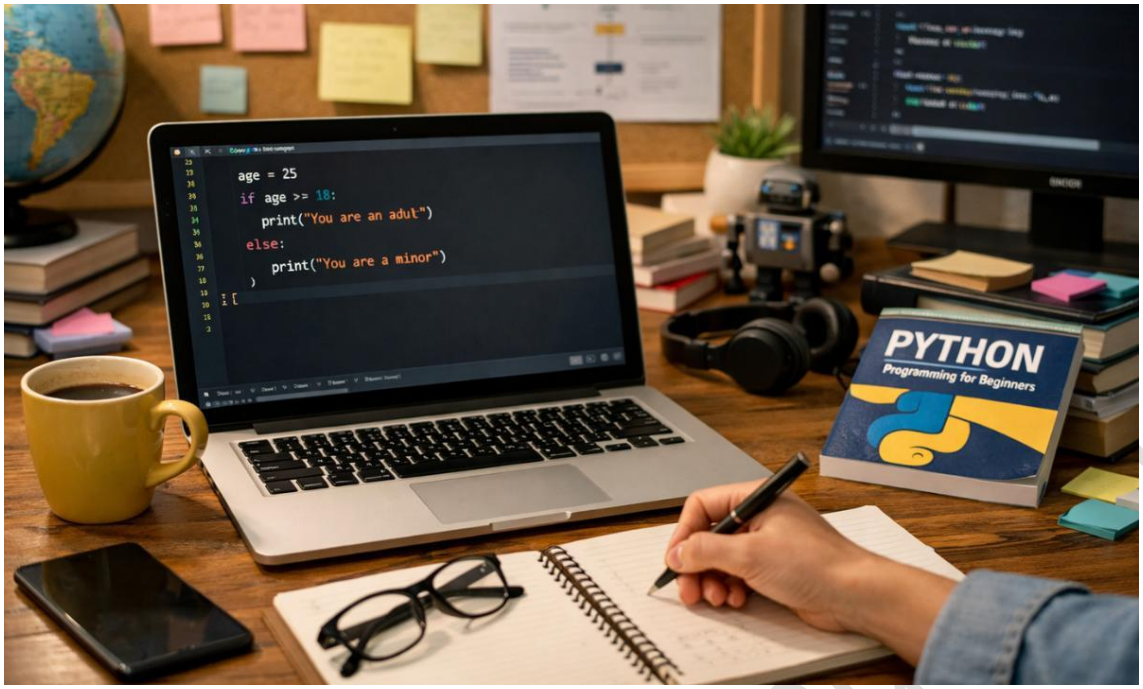
Programmers use algorithms to plan what a computer should do before writing code.

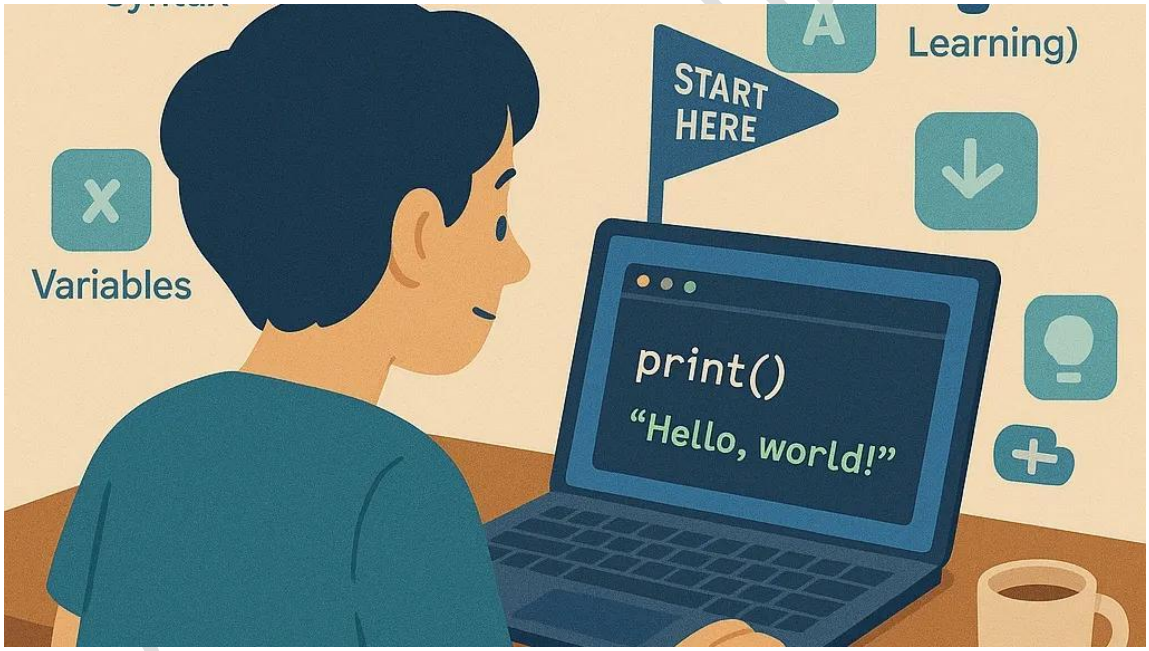
4. PROGRAMMING LANGUAGES & PYTHON

THEORY



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What is a Programming Language?

Tell students:

Humans communicate using languages such as English, Hindi, Telugu, etc.
Computers need instructions written in programming languages.

Examples:

- Python
- Java
- JavaScript
- C
- C++
- C#
- Go
- Swift

WHAT IS PYTHON?

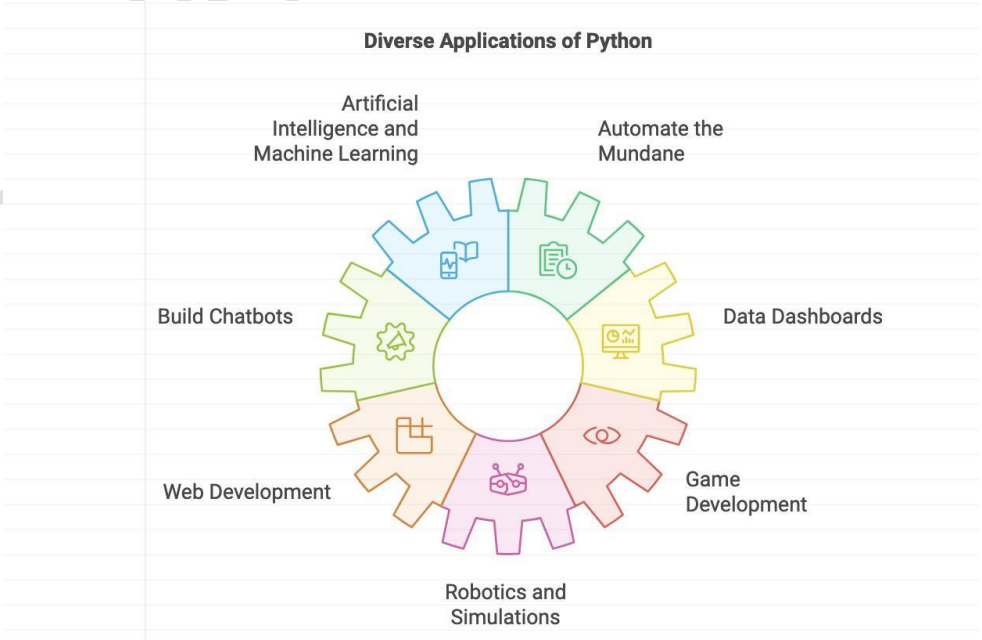
Definition

Python is a high-level programming language used to create software, automate tasks, analyze data, build AI systems, and much more.

Why are we learning Python?

Because Python has:

- Simple and readable syntax
- Huge ecosystem
- Applications in AI
- Applications in robotics
- Applications in data science
- Applications in automation
- Applications in web development



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Applications of Python



5. FIRST PYTHON PROGRAM

PRACTICAL

Now students open their Python environment.

Tell them:

"This is your first Python program. Type it exactly."

```
print("Hello, World!")
```

Run it.

Output:

```
Hello, World!
```

Then explain:

```
print()
```

Used to Show message on the screen.

PRACTICAL ACTIVITY 1

Students type:

```
print("Hello!")
print("My name is _____.")
print("I am learning Python.")
print("I want to become an engineer.")
```

Expected Output:

```
Hello!
My name is Rahul.
I am learning Python.
I want to become an engineer.
```

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QUICK CHALLENGE

Give students a short break.

After the break, give them this challenge:

Make Python display:

```
My name is ______
My school is ______
My favorite subject is ______
My favorite technology is ______
I want to become ______
```

6. PYTHON print() & BASIC SYNTAX

PRACTICAL

Explain the difference between text and numbers.

Text

```
print("Python")
print("Engineering")
print("Robot")
```

Text is written inside quotation marks.

Numbers

```
print(10)
print(25)
print(100)
```

Numbers don't need quotation marks when we want Python to treat them as numbers.

PRACTICAL ACTIVITY 2 — CALCULATIONS

Addition

```
print(10 + 20)
```

Output:

```
30
```

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Subtraction

```
print(50 - 20)
```

Output:

```
30
```

Multiplication

```
print(5 * 6)
```

Output:

```
30
```

Division

```
print(20 / 4)
```

Output:

```
5.0
```

MINI CODING CHALLENGE

calculate:

```
print(25 + 75)
print(100 - 45)
print(8 * 7)
print(81 / 9)
```

Outputs

```
100
55
56
9.0
```

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7. MINI PROJECT — "ABOUT ME"

Students create their first small Python project.

Project

```
print("==============================")
print("       ABOUT ME")
print("==============================")

print("My name is Alex")
print("I am 12 years old")
print("My favorite subject is Science")
print("My favorite technology is Robotics")
print("I want to become a Robotics Engineer")

print("==============================")
print("I LOVE PYTHON!")
print("==============================")
```

Students replace the example information with their own.

LEVEL 2 CHALLENGE

Students add calculations:

```
print("My age is 12")
print("In 5 years I will be:")
print(12 + 5)
```

Output:

```
My age is 12
In 5 years I will be:
17
```

8. CODING GAME — FIND THE MISTAKE

Show students:

```
print(Hello Python)
```

Correct version:

```
print("Hello Python")
```

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Mistake 1

```
Print("Hello")
```

Correct:

```
print("Hello")
```

Explain:

Python is case-sensitive. print and Print are different.

Mistake 2

```
print("Hello Python"
```

find the missing character.

Correct:

```
print("Hello Python")
```

9. FINAL CLASS PROJECT

"MY FIRST PYTHON PROGRAM"

Every student should create:

```
print("===== MY FIRST PYTHON PROGRAM =====")

print("Hello!")
print("My name is _____")
print("I am learning Python")
print("I love technology")
print("My dream is to become _____")

print("Here is my calculation:")
print(20 + 30)

print("I am a Future Programmer!")
```

Success Criteria

Students should have: At least 5 print() statements
Their name
Their career goal
One calculation

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Correct quotation marks
Correct brackets

10. QUICK QUIZ

Q1. What is programming?

Answer: Giving instructions to a computer.

Q2. What is Python?

Answer: A programming language.

Q3. What does `print()` do?

Answer: Displays information on the screen.

Q4. What will this produce?

```
print(10 + 5)
```

Answer: `15`

Q5. Which is correct?

A.

```
Print("Hello")
```

B.

```
print("Hello")
```

Answer: B

Learning Outcomes

By the end of Class 1, students will be able to:

1. Define programming and explain its purpose in everyday technology.
2. Explain how computers follow instructions to perform tasks.
3. Understand the concept of an algorithm as a step-by-step solution to a problem.
4. Identify common programming languages and explain what Python is.
5. Describe some real-world applications of Python, including AI, robotics, automation, data science, and web development.

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6. Understand the basic structure of a Python program.
7. Use the Python **print() function** to display text and numbers.
8. Perform basic arithmetic calculations using Python.
9. Identify and correct simple Python syntax mistakes.
10. Create a simple **"About Me / My First Python Program"** using print().

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